

## Required Acoustic Transmitter Specifications

The transmitters must use a binary phase shift keyed (BPSK) code pulse to achieve a unique code.

Transmitters must be the smallest acoustic tag available that meets the life and output requirement of the JSATS Program.

Transmitters must contain a 4-place alphanumeric code ID

Transmitters must operate between 0 and 31 degrees Celsius.

Transmitters must emit a frequency within 416.7 kHz  $\pm$  0.5%.

Transmitters must have a pulse rate interval (PRI), code ID and duty cycling that is configurable by the user with an ATS brand Pinger Dish.

Transmitters must contain an 8-bit Barker code, 16-bit tag ID code and 8-bit cyclic redundancy check (CRC).

## Required Acoustic Receiver Specifications

The receivers must detect, decode, and store to memory all valid tag codes from binary phase-shift keyed (BPSK) acoustic signals on the Juvenile Salmonid Acoustic Telemetry System (JSATS) architecture.

Receivers must be cabled- or shore-based where the electronics are housed above water in a NEMA type 4x enclosure and connect to a hydrophone placed underwater via a 100 ft cable.

Uses 12 volt external power supply

Direct USB connection to receiver allows user to set receiver ID, clock synchronization, access to stored configuration, and viewing of real-time data feed via vendor provided software that provides a graphical user interface.

RS-232 data offload via cellular modem or other remote communication method (user supplied).

Removable 32 GB SDHC memory card for storage of data.

Time is continuously set and corrected for drift via GPS.

System voltage and external temperature are recorded in data file.

Can decode JSATS transmitters within 416.7 kHz  $\pm$  0.5%.

Hydrophone sensitivity of -174 dB re 1uPa/V at 1 m.

Horizontal plane deviation of 4dB or less.

Vertical plane deviation of 4dB or less,  $\pm 130$  degrees from vertical.

Operating temperature range of 0-50 °C.

Parts and labor warranty of 1 year or greater.